Chungggi **Lee**

PhD Candidate @ Harvard University

Science and Engineering Complex, 150 Western Ave, Boston, MA

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Education

Harvard University *Boston, MA*

Ph.D in Computer Science *Sep. 2023 - Now*

- GPA: 3.92/4.0, Advisor : Prof. Hanspeter Pfister

Ulsan National Institute of Science and Technology *Ulsan, South Korea*

M.S in Electrical and Computer Engineering *Mar. 2018 - Feb. 2020*

- GPA : 4.17/4.3, Advisor : Prof. Sungahn Ko
- Thesis Title: GUIComp: A GUI Design Assistantwith RealTime, MultiFaceted Feedback

Ulsan National Institute of Science and Technology *Ulsan, South Korea*

B.S in Electrical and Computer Engineering *Mar. 2014 - Feb. 2018*

- Magna Cum Laude
- Thesis Title: A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion.

Publications

International

P1. **Chunggi Lee**, Tica Lin, Yalong Yang, Hanspeter Pfister, **"Who's That Player?: Externalizing Query Interpretation in Spoken XR Sports Interaction"**, IEEE Transactions on Visualization and Computer Graphics [PDF]

P2. **Chunggi Lee**, Seonwook Park, Wanhua Li, Umar Iqbal*, Hanspeter Pfister*, **"DETRAM: End-to-end DEtection, Tracking and Recovery of HumAn Meshes"**, European Conference on Computer Vision [PDF]

P3. **Chunggi Lee***, Hayato Saiki*, Tica Lin, Eiji Ikeda, Kenji Suzuki, Chen Zhu-Tian, Hanspeter Pfister, **"ViSTAR: Virtual Skill Training with Augmented Reality with 3D Avatars and LLM coaching agent"**, ACM CHI Conference on Human Factors in Computing Systems [PDF]

P4. Hayato Saiki, **Chunggi Lee**, Hikari Takahashi, Tica Lin, Hidetada Kishi, Kaori Tachibana Ibaraki, Yasuhiro Suzuki, Hanspeter Pfister, Kenji Suzuki, **"BRIDGE: Borderless Reconfiguration for Inclusive and Diverse Gameplay Experience via Embodiment Transformation."**, ACM CHI Conference on Human Factors in Computing Systems **(Best Paper Award)** [PDF] NEWS1 NEWS2

P5. **Chunggi Lee**, Ut Gong, Tica Lin, Stefanie Zollmann, Scott A Epsley, Adam Petway, Hanspeter Pfister, **"VAIR: Visual Analytics for Injury Risk Exploration in Sports."**, IEEE Transactions on Visualization and Computer Graphics (**IEEE VIS VAHC Workshop**), 2025.

P6. **Chunggi Lee**, Tica Lin, Hanspeter Pfister, Zhu-Tian Chen, **"Sportify: Question Answering with Embedded Visualizations and Personified Narratives for Sports Video."**, IEEE Transactions on Visualization and Computer Graphics (**IEEE VIS**), 2024. [PDF] [Webpage]

P7. Namhyuk Ahn, Junsoo Lee, **Chunggi Lee**, Kunhee Kim, Daesik Kim, Seung-Hun Nam, Kibeom Hong, **"DreamStyler: Paint by Style Inversion with Text-to-Image Diffusion Models."**, The Association for the Advancement of Artificial Intelligence (**AAAI**), 2024. [PDF] [Webpage] [Code]

P8. Han Kim*, **Chunggi Lee***, Junsoo Lee*, Dohyun Kim, Kwangjin Lee, Moohyun Oh, Daesik Kim, **"FlatGAN: A Holistic Approach for Robust Flat-Coloring in High-Definition with Understanding Line Discontinuity."**, ACM Multimedia (**MM**), 2023. PDF

P9. Cholmin Kang, **Chunggi Lee**, Heon Song, Minuk Ma, Sergio Pereira, **"Variability Matters : Evaluating inter-rater variability in histopathology for robust cell detection."**, European Conference on Computer Vision Workshop On AI-Enabled Medical Image Analysis (**ECCVW**), 2022. [PDF]

P10. **Chunggi Lee**, Seonwook Park, Heon Song, Jeongun Ryu, Sanghoon Kim, Haejoon Kim, Sergio Pereira, Donggeun Yoo, **"Interactive Multi-Class Tiny-Object Detection."**, IEEE Conference on Computer Vision and Pattern Recognition (**CVPR**), 2022. [PDF] [VIDEO]

P11. Cheonbok Park, **Chunggi Lee**, Hyojin Bahng, Yunwon Tae, Kihwan Kim, Seungmin Jin, Sungahn Ko, Jaegul Choo, **"ST-GRAT: A novel spatio-temporal graph attention networks for accurately forecasting dynamically changing road speed."**, ACM International Conference on Information and Knowledge Management (**CIKM**), 2020 (20.9% acceptance rate). [PDF]

P12. **Chunggi Lee**, Sanghoon Kim, Dongyun Han, Hongjun Yang, Young-Woo Park, Bum Chul Kwon, Sungahn Ko, **"GUIComp: A GUI Design Assistantwith RealTime, MultiFaceted Feedback"**, ACM CHI Conference on Human Factors in Computing Systems (**CHI**), 2020, Accepted. [PDF] [Preview Video] [VIDEO]

P13. **Chunggi Lee**, Yeonjun Kim, Seungmin Jin, Dongmin Kim, Ross Maciejewski, David Ebert, and Sungahn Ko, **"A Visual Analytics System for Exploring, Monitoring, and Forecasting Road Traffic Congestion."** IEEE transactions on visualization and computer graphics (**TVCG** IF=4.579), 2019 (Proc. **IEEE VIS'19**), Accepted. [LINK] [PDF] [VIDEO][NEWS1] [NEWS2] [NEWS3]

Preprint, Poster, Domestic

D1. Hyunwook Lee, **Chunggi Lee**, Hongkyu Lim, Sungahn Ko, **"TILDE-Q: A Transformation Invariant Loss Function for Time-Series Forecasting."**, (Under Review), 2022.

D2. Juyoung Oh, **Chunggi Lee**, Hwiyeon Kim, Kihwan Kim, Osang Kwon, Eric D. Ragan, Bum Chul Kwon, Sungahn Ko, **"An Empirical Study on the Relationship Between the Number of Coordinated Views and Visual Analysis."**, Arxiv. [PDF]

D3. **Chunggi Lee**, Juyoung Oh, Seungmin Jin, Isaac Cho, and Sungahn Ko, **"A Graphical Workflow Exploration Environment For Visual Analytics."**, Arxiv. [PDF]

D4. Kihwan Kim, Sanghoon Kim, **Chunggi Lee**, Sungahn Ko, **"Modeling Exploration/Exploitation Decisions through Mobile Sensing for Understanding Mechanisms of Addiction."**, **(MobiSys)**, 2019 (Poster)

D5. J. Lee, K. Kim, **C. Lee**, and S. Ko, "Visualization based Deep Learning Analysis Technology." The Korean Society for Noise and Vibration Engineering **(KSNVE)**, 2017, Accepted. [LINK] [PDF]

D6. Y. Oh, **C. Lee**, J. Oh, J. Yang, H. Gwag, S. Moon, S. Park, and S. Ko, "Introdcution of Visual Analytics." **Korea Computer Graphics Society**, 2016, Accepted. [LINK] [PDF]

Professional & Research Experience

Dolby Laboratories. — *Atlanta, Goergia*

Research Scientist Intern[LINK] — *May 2026 - Aug 2026*

- Spatial audio system and model for personalization

NVIDIA. — *Remote*

Research Collaboration[LINK] — *March 2025 - March 2026*

- End-to-end Detection, Tracking and Recovery of Human Meshes

Naver Webtoon AI. — *Gyeonggi-do, South Korea*

Research Scientist (Alternative Military Service) [LINK] — *Aug 2022 - Aug 2023*

- Conducted research on **stable diffusion models** to support the process of webtoon (comics) creation
- Designed and built an **AI platform** frontend and backend to distribute deep learning models for users.

Lunit Inc. — *Seoul, South Korea*

AI Research Engineer (Alternative Military Service) [LINK] — *Mar 2020 - Aug-2022*

- Designed and built interactive deep learning models for reducing cost of annotations **(e.g., object detection and segmentation)**. (P2)
- Implemented an annotation tool and an internal operation tool with **Django, DRF, React, and mui.**
- Deployed and served deep learning models by using **TorchServe and Fast API.**
- Analyzed quality of pathology data which have significant different among experts and proposed a new method for evaluating pathology large-scale dataset (P1)

Interactive Visual Analysis & Data Exploration Research Lab — *Ulsan, South Korea*

Researcher & Developer [LINK] — *Jun 2016 - Mar 2020*

- Built interactive tools and visual analytic systems for **mobile GUI design authoring for novice** (P4), **traffic congestion forecasting and propagation** (P5), **multiple coordinate visualization** (D3), and **gene expression** (D4).
- Implemented **spatio-temporal deep learning models** (P3), mobile GUI recommendation deep learning, biomedical data clustering

Honors & Awards

2025	**Best Paper Award (Top 1%)**, ACM CHI	*Barcelona, Spain*
2017	**4th Place & Bronze Prize**, National Super Computing Competition	*Ulsan, South Korea & KISTI & UNIST*
2017	**3rd Place & Bronze Prize**, Naver UNIST Undergraduate Poster Award '17 UNIST $1,500 [POSTER]	*Naver & UNIST*
2016 & 2017	**Semester GPA 3.94/4.3, GPA 3.93/4.3**, Semester Academic Excellence Award for 2016 Fall Semester, 2017 Fall Semester	*UNIST*

Patent

- **Graphic User Interface Assistant Device and Method in Mobile Environment** – Application Number : 10-2019-0001190
- **Traffic Information Visualization Analysis Device and Method** – Application Number : 10-2018-0079727 (Acquired - $0.1M)

Services

	Reviewer, ACM CHI (2024, 26), IEEE VIS (2023-26), CVPR (2026), PacificVis (2026), TVCG(2026), DIS (2026), TGRS (2026), ECCV (2026)	
2019, 24	**Student Volunteer**, IEEE VIS	*Vancouver, Canada*
2015, 19	**Teaching Assistant**, Object Oriented Programming	*Ulsan, South Korea*

Mentoring

2025 **Mentee**, **Hayato Saiki** (University of Tsukuba), **Nina Doerr** (University of Stuttgart), **Max Wagner (HCRP Award)** (Harvard University), **Alexandre Ben Ahmed Kontoul (6.0/6.0)** (EPFL -> Amazon) *Boston, MA*

2024 **Mentee**, **Arthur Andre - MS Thesis** (EPFL), **Zhuo Wang** (Xi'an Jiaotong-Liverpool University), **Ut(Jo Jo) Gong** (University of Washington -> Columbia) *Boston, MA*